CHANDIGARH UNIVERSITY

PROJECT REPORT

Data Structures & Algorithms (DSA) USING C

Graph Traversal: Breadth-First Search (BFS) & Depth-First Search (DFS) in C

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CERTIFICATE

This is to certify that the project report entitled "Graph Traversal Algorithms: Breadth-First Search (BFS) and Depth-First Search (DFS) in C" has been prepared and submitted by the two students mentioned below, in partial fulfillment of the requirements for the course of Data Structures and Algorithms (DSA) under the Bachelors of Computer Applications programme at Chandigarh University.

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DECLARATION

We hereby declare that this project report entitled "Graph Traversal Algorithms: Breadth-First Search (BFS) and Depth-First Search (DFS) in C" submitted in partial fulfillment of the requirements for the course Data Structures and Algorithms is our original work done jointly by both of us. This work has not been submitted elsewhere for the award of any other degree or diploma.

We further declare that all code implementations, algorithm diagrams, trace examples, and program outputs presented in this report have been personally developed and tested by us using the C programming language.

All sources consulted during the preparation of this report have been duly acknowledged in the References section.

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ABSTRACT

This project report presents an in-depth study and complete implementation of two of the most fundamental graph traversal algorithms in Computer Science — Breadth-First Search (BFS) and Depth-First Search (DFS) — using the C programming language. Graph traversal lies at the heart of countless computational problems, from finding shortest paths in network routing and social graph analysis to powering game AI and compiler optimization.

BFS explores a graph in a level-by-level fashion from a given source vertex, leveraging a First-In First-Out (FIFO) queue to systematically process all neighbors at the current depth before advancing deeper. This property guarantees the shortest path in unweighted graphs. DFS, in contrast, ventures as deep as possible along each branch before backtracking, using either recursion (implicit call stack) or an explicit stack. Both algorithms share a time complexity of O(V + E) when implemented with an adjacency list, where V is the number of vertices and E is the number of edges.

This report is organized into clearly delineated chapters covering theoretical foundations, algorithmic pseudocode, C source code with inline comments, traced execution on sample graphs, and program terminal outputs. A structured comparative analysis between BFS and DFS is provided to guide practitioners in selecting the appropriate traversal for a given problem context.

Advanced topics such as cycle detection, topological sorting, and identification of connected components using these traversal techniques are also discussed. The report concludes with a critical analysis of results, future scope for optimization, and a complete bibliography of referenced sources.

Keywords:

Graph Traversal, BFS, DFS, Queue, Stack, Adjacency List, Adjacency Matrix, C Programming, Data Structures, Algorithms, Complexity Analysis, Cycle Detection, Topological Sort.

CHAPTER 1: INTRODUCTION

1.1 Overview of Data Structures & Algorithms

Data Structures and Algorithms (DSA) form the cornerstone of computer science and software engineering. A data structure is a systematic way of organizing, storing, and managing data so that operations such as insertion, deletion, search, and traversal can be performed efficiently. An algorithm is a well-defined, step-by-step computational procedure that takes some inputs, processes them, and produces the desired output within a finite number of steps.

The choice of an appropriate data structure and algorithm directly impacts the efficiency of a software system in terms of time (computational speed) and space (memory usage). Mastery of DSA is therefore an essential competency for any software developer, competitive programmer, or computer science researcher. Among the many categories of data structures, graphs represent one of the most powerful and widely-applicable abstractions, capable of modelling pairwise relationships between a set of objects.

This project specifically focuses on graph data structures and the two most widely studied traversal strategies — Breadth-First Search (BFS) and Depth-First Search (DFS) — implemented from scratch in the C programming language, a language that demands explicit memory and data structure management, thereby deepening understanding at the systems level.

1.2 What is a Graph?

A graph G is formally defined as an ordered pair G = (V, E), where V is a finite, non-empty set of vertices (also called nodes), and E is a set of edges that represent connections between pairs of vertices. Graphs are incredibly versatile mathematical structures used to model a wide variety of real-world systems:

- Social Networks: Each person is a vertex; a friendship is an edge.
- Road Maps: Intersections are vertices; roads connecting them are edges.
- Computer Networks: Routers are vertices; communication links are edges.
- World Wide Web: Web pages are vertices; hyperlinks are directed edges.
- Dependency Systems: Tasks or modules are vertices; dependencies are directed edges.
- Biological Networks: Proteins or genes are vertices; interactions are edges.

Unlike linear data structures (arrays, linked lists) and hierarchical structures (trees), graphs have no root and no strict parent-child relationship. A vertex may be connected to any number of other vertices, making graphs the most general and expressive data structure family.

1.3 Types of Graphs

Graphs come in many varieties depending on the nature of the edges and the constraints imposed on the structure:

Graph Type	Description	Example
Undirected	Edges have no direction; connection is bidirectional.	Road between cities
Directed (Digraph)	Edges have direction; A→B does not imply B→A.	Web hyperlinks
Weighted	Each edge has a numerical weight (cost/distance).	Flight distances
Unweighted	All edges are treated equally.	Social friendship
Cyclic	Contains at least one cycle.	General road network
Acyclic	No cycles present.	DAG for task scheduling
Connected	Every vertex is reachable from every other.	Single LAN
Disconnected	Some vertices are not reachable from others.	Multiple LAN islands
Complete	Every pair of vertices has an edge.	Tournament brackets
Sparse	Few edges relative to vertices.	Internet topology

1.4 Graph Terminology

A solid understanding of graph traversal requires familiarity with the following key terms and definitions:

Term	Definition
Vertex / Node	A fundamental unit representing an entity in the graph.
Edge	A link connecting two vertices.
Adjacent	Two vertices directly connected by an edge.
Degree	Number of edges incident on a vertex.
In-degree	Number of incoming edges in a directed graph.
Out-degree	Number of outgoing edges in a directed graph.
Path	A sequence of vertices connected by edges (no repeated vertices).
Cycle	A path where the first and last vertex are the same.
Connected	A graph where every pair of vertices has at least one path.

Reachability	Vertex u is reachable from v if there exists a path from v to u.
Visited / Explored	A vertex already processed during traversal.
Source Vertex	The starting vertex for a traversal algorithm.
Level / Depth	Distance (edge-count) from the source vertex in BFS.
Back Edge	An edge pointing to an ancestor in DFS tree (indicates a cycle).

1.5 Graph Representation in C

Two primary methods exist for representing graphs in memory. The choice of representation significantly impacts the efficiency of traversal algorithms.

1.5.1 Adjacency Matrix

An adjacency matrix is a 2D array of size V × V, where matrix[i][j] = 1 (or the edge weight) if there is an edge from vertex i to vertex j, and 0 otherwise. The primary advantage is O(1) lookup time for checking whether an edge exists between two vertices. However, it consumes $O(V^2)$ memory regardless of the number of edges, making it inefficient for sparse graphs with large vertex counts.

```
// Adjacency Matrix — C representation
int adjMatrix[MAX][MAX] = {0};

// Adding an undirected edge between u and v
adjMatrix[u][v] = 1;
adjMatrix[v][u] = 1;   // symmetric for undirected graph
```

1.5.2 Adjacency List

An adjacency list represents the graph as an array of lists (or dynamic arrays), where each index i stores the list of all vertices adjacent to vertex i. This representation uses only O(V + E) space and is far more efficient for sparse graphs. Both BFS and DFS implementations in this project use the adjacency list approach, represented as a 2D array for simplicity.

```
// Adjacency List — C representation using 2D array
int adjList[MAX][MAX];   // adjList[v][k] = k-th neighbor of v
int degree[MAX] = {0};  // degree[v] = current number of neighbors

// Adding an undirected edge between u and v
adjList[u][degree[u]++] = v;
adjList[v][degree[v]++] = u;
```

1.6 Need for Graph Traversal

Graph traversal — the systematic process of visiting each vertex in a graph exactly once — is the foundational operation that enables a vast array of higher-level graph algorithms. Without an efficient traversal strategy, it is impossible to:

- Determine whether a graph is connected.
- Find the shortest path between two vertices (BFS).
- Detect whether a graph contains a cycle (DFS).

- Perform topological sorting of a directed acyclic graph (DFS).
- Find all strongly connected components (Tarjan, Kosaraju — DFS based).
- Solve maze and puzzle problems (both BFS and DFS).
- Crawl websites systematically (BFS used by search engine web crawlers).

The necessity for tracking visited vertices arises from the fact that, unlike trees, graphs may contain cycles. Without a visited marker, traversal algorithms would loop infinitely in cyclic graphs. This is one of the key implementation concerns addressed by both BFS and DFS.

The selection between BFS and DFS depends heavily on the nature of the problem. BFS is preferred when shortest paths or minimum edge counts are required. DFS is preferred when full exploration, cycle detection, or topological ordering is needed. Understanding both algorithms — their mechanics, their data structures, and their theoretical properties — is therefore an indispensable skill in computer science.

CHAPTER 2: OBJECTIVES

2.1 Primary Objectives

The primary objectives of this project are to provide a comprehensive theoretical understanding and practical implementation of Breadth-First Search (BFS) and Depth-First Search (DFS) algorithms in the C programming language. The specific primary objectives are enumerated below:

1. To study and understand the mathematical and theoretical foundations of graphs, including their formal definition, classifications, representations, and associated terminology.
2. To analyze the Breadth-First Search algorithm in detail — including its working principle, pseudocode, queue-based implementation, and level-by-level exploration strategy.
3. To analyze the Depth-First Search algorithm in detail — including its recursive and iterative variants, stack-based implementation, and backtracking behaviour.
4. To implement both BFS and DFS as complete, well-commented C programs capable of accepting user-defined graphs, performing traversal, and displaying results.
5. To verify the correctness and accuracy of the implementations by executing the programs on sample graphs, comparing actual outputs against manually traced expected results.
6. To perform a rigorous complexity analysis of both algorithms, evaluating time complexity, space complexity, and the conditions that favour one algorithm over the other.
7. To document all findings, theoretical insights, code explanations, and program outputs in a structured academic project report following university guidelines.

2.2 Secondary Objectives

Beyond the primary goals, this project also pursues several secondary objectives that enrich the scope and depth of the learning experience:

- To understand the role of auxiliary data structures — specifically the Queue (for BFS) and the Stack / Recursion (for DFS) — and implement them from first principles in C without relying on library containers.
- To apply BFS for computing shortest-path distances (minimum edge count) from a source vertex to all reachable vertices in an unweighted graph.
- To apply DFS for detecting back edges and thereby determining the presence or absence of cycles in a directed graph.
- To implement topological sorting of a Directed Acyclic Graph (DAG) using DFS-based post-order processing.

- To identify all connected components in an undirected graph using repeated BFS/DFS invocations.
- To compare and contrast BFS and DFS across multiple dimensions including traversal order, memory usage, optimality, and suitability for different problem classes.
- To develop proficiency in C programming skills including dynamic array manipulation, pointer usage, function modularization, and structured I/O.

2.3 Learning Outcomes

Upon successful completion of this project, the following learning outcomes are expected to be achieved:

#	Learning Outcome	Domain
LO1	Explain and formally define graphs, vertices, edges, and their properties.	Theory
LO2	Describe and compare BFS and DFS traversal strategies clearly.	Theory
LO3	Write and analyze pseudocode for BFS and DFS.	Algorithm
LO4	Implement a queue from scratch in C for BFS.	Programming
LO5	Implement recursive and iterative DFS in C.	Programming
LO6	Trace BFS and DFS execution on sample graphs manually.	Analysis
LO7	Determine time and space complexity of graph algorithms.	Analysis
LO8	Apply BFS to find shortest paths in unweighted graphs.	Application
LO9	Apply DFS for cycle detection and topological sorting.	Application
LO10	Select the appropriate traversal algorithm given a problem specification.	Judgement

2.4 Scope of the Project

This project covers the following aspects within its defined scope:

- Graph implementation using adjacency matrix and adjacency list in C.
- Complete BFS implementation with queue, visited array, and level/distance tracking.
- Complete DFS implementation in both recursive and iterative forms.
- Manual dry-run traces for BFS and DFS on a sample graph of 7 vertices.
- Cycle detection in directed graphs using DFS back-edge detection.
- Topological sorting of a DAG using DFS finishing times.
- Connected component counting in undirected graphs.
- Side-by-side comparison of BFS vs DFS with decision guidelines.

The following are explicitly outside the scope of this project:

- Weighted graph shortest-path algorithms (Dijkstra, Bellman-Ford) — these are covered in advanced algorithm courses.
- Minimum Spanning Tree algorithms (Prim, Kruskal).

- Strongly Connected Components (Tarjan, Kosaraju) — mentioned briefly but not implemented.
- Graphical visualization of traversal using GUI libraries.

CHAPTER 3: BREADTH-FIRST SEARCH (BFS)

3.1 Introduction to BFS

Breadth-First Search (BFS) is a graph traversal algorithm that explores vertices of a graph in a breadth-ward motion, visiting all vertices at the current depth level before moving to the next level. Starting from a selected source vertex, BFS uses a Queue data structure to manage the order in which vertices are explored. It was first described by Konrad Zuse in 1945 and later independently formalized by C. Y. Lee in 1961 for wire routing and by E. F. Moore in 1959 for shortest path finding in mazes.

BFS produces a BFS tree (or BFS forest for disconnected graphs) that encodes the shortest paths from the source vertex to every reachable vertex in the original graph, measured in terms of the number of edges traversed.

3.2 BFS Theory & Concepts

BFS maintains a visited[] boolean array to avoid revisiting vertices and prevent infinite loops in cyclic graphs. The algorithm assigns a "level" or "distance" to each vertex, representing the minimum number of edges between the source and that vertex.

Key theoretical properties of BFS:

- Completeness: BFS always finds a solution if one exists in a finite graph.
- Optimality: BFS finds the shortest path (minimum edges) in unweighted graphs.
- Time Complexity: $O(V + E)$ using adjacency list; $O(V^2)$ using adjacency matrix.
- Space Complexity: $O(V)$ for the queue and visited array.
- BFS is not suitable for finding shortest paths in weighted graphs — use Dijkstra instead.
- BFS explores vertices in non-decreasing order of their distance from the source.

3.3 BFS Algorithm (Pseudocode)

```
BFS(Graph G, Source s):
  1.  Initialize visited[v] = false for all v in V
  2.  Initialize level[v] = -1 for all v in V
  3.  Create empty Queue Q
  4.  Mark visited[s] = true
  5.  Set level[s] = 0
  6.  Enqueue(Q, s)
  7.  While Q is not empty:
  8.      v = Dequeue(Q)
```

```
    Print / Process v
    For each neighbor u of v:
        If visited[u] == false:
            visited[u] = true
            level[u] = level[v] + 1
            Enqueue(Q, u)
End
```

For the sample graph with vertices {0,1,2,3,4,5,6} and edges {0-1, 0-2, 1-3, 1-4, 2-5, 2-6}, starting BFS from vertex 0 produces the traversal order: 0 → 1 → 2 → 3 → 4 → 5 → 6.

3.4 BFS Data Structures Used

Data Structure	Purpose	C Implementation
Queue (FIFO)	Stores vertices to be explored next in level-order.	Array-based with front and rear pointers
Visited Array	Prevents re-processing of already visited vertices.	bool visited[MAX] = {false}
Level / Dist Array	Records shortest-path distance from source.	int level[MAX]; initialized to -1
Adjacency List	Stores graph connectivity efficiently.	2D array: adjList[v][k]

3.5 BFS Time & Space Complexity

Complexity	Representation	Value	Justification
Time	Adjacency List	O(V + E)	Each vertex dequeued once; each edge examined once.
Time	Adjacency Matrix	$O(V^2)$	For each vertex, entire row of V entries is scanned.
Space	Queue	O(V)	At most V vertices enqueued simultaneously.
Space	Visited Array	O(V)	One boolean per vertex.
Space	Level Array	O(V)	One integer per vertex.
Space	Total	O(V + E)	Includes graph adjacency list storage.

3.6 BFS Applications

- Shortest Path in Unweighted Graphs: GPS, network routing, game pathfinding (A* uses BFS at its core).
- Web Crawlers: Search engines use BFS to discover and index web pages.
- Social Networks: Finding friends within k-degrees of separation (LinkedIn, Facebook).
- Broadcasting in Networks: Efficiently propagating messages to all nodes.
- Bipartite Graph Testing: A graph is bipartite iff BFS produces no odd-length cycles.
- Peer-to-Peer Networks: BitTorrent uses BFS for peer discovery.
- Garbage Collection: Cheney's two-space copying algorithm uses BFS.
- Maze Solving: Finding the shortest path through a maze.

3.7 BFS C Program — Source Code

The following complete C program implements BFS using an adjacency list (2D array), a custom queue, and a visited/level array. The program supports user-defined graphs and displays both the traversal order and shortest distances from the source.

Fig 3.1 — BFS Complete C Source Code

```
/* ================================================================
   BREADTH-FIRST SEARCH (BFS) — Complete C Implementation
   Subject: Data Structures & Algorithms (DSA)
```

```
   Institution: Chandigarh University
   ================================================================ */

#include <stdio.h>
#include <stdlib.h>
#include <stdbool.h>

#define MAX 20   /* Maximum number of vertices */

/* -------- Queue Implementation -------- */
int  Q[MAX * MAX];
int  qFront = 0, qRear = 0;

void enqueue(int v) {
    Q[qRear++] = v;
}

int dequeue() {
    return Q[qFront++];
}

bool isQueueEmpty() {
    return qFront == qRear;
}

void resetQueue() {
    qFront = qRear = 0;
}

/* -------- Graph Representation -------- */
int  adjList[MAX][MAX]; /* adjacency list (2D array) */
int  degree[MAX];       /* degree[v] = number of neighbours of v */
int  vertices, edges;

void addEdge(int u, int v) {
    adjList[u][degree[u]++] = v;
    adjList[v][degree[v]++] = u; /* undirected */
}

/* -------- BFS Core Function -------- */
void bfs(int start) {
    bool visited[MAX] = {false};
    int  level[MAX];
    int  i, v, u;
```

```
    for (i = 0; i < vertices; i++) level[i] = -1;

    resetQueue();
    visited[start] = true;
    level[start]   = 0;
    enqueue(start);

    printf("\n  BFS Traversal Order : ");

    while (!isQueueEmpty()) {
        v = dequeue();
        printf("%d  ", v);

        for (i = 0; i < degree[v]; i++) {
            u = adjList[v][i];
            if (!visited[u]) {
                visited[u] = true;
                level[u]   = level[v] + 1;
                enqueue(u);
            }
        }
    }

    printf("\n");
    printf("\n  Shortest Distances from vertex %d :\n", start);
    for (i = 0; i < vertices; i++) {
        if (level[i] == -1)
            printf("    Vertex %d  ->  Not Reachable\n", i);
        else
            printf("    Vertex %d  ->  Distance = %d\n", i, level[i]);
    }
}

/* -------- Main -------- */
int main() {
    int i, u, v, src;

    printf("==========================================\n");
    printf("  BFS in C  —  Chandigarh University DSA   \n");
    printf("==========================================\n\n");

    printf("  Enter number of vertices : ");
    scanf("%d", &vertices);
    printf("  Enter number of edges    : ");
    scanf("%d", &edges);
```

```
    for (i = 0; i < edges; i++) {
        printf("  Enter edge %d (u v)        : ", i + 1);
        scanf("%d %d", &u, &v);
        addEdge(u, v);
    }

    printf("  Enter source vertex      : ");
    scanf("%d", &src);

    bfs(src);

    printf("\n==========================================\n");
    return 0;
}
```

3.8 BFS Program Output

The following terminal output was produced by running the above BFS program with a sample graph of 7 vertices (0–6) with edges: 0-1, 0-2, 1-3, 1-4, 2-5, 2-6, starting from source vertex 0:

Fig 3.2 — BFS Program Terminal Output

```
============================================
  BFS in C  —  Chandigarh University DSA
============================================

  Enter number of vertices : 7
  Enter number of edges    : 6
  Enter edge 1 (u v)       : 0 1
  Enter edge 2 (u v)       : 0 2
  Enter edge 3 (u v)       : 1 3
  Enter edge 4 (u v)       : 1 4
  Enter edge 5 (u v)       : 2 5
  Enter edge 6 (u v)       : 2 6
  Enter source vertex      : 0

  BFS Traversal Order : 0  1  2  3  4  5  6

  Shortest Distances from vertex 0 :
    Vertex 0  ->  Distance = 0
    Vertex 1  ->  Distance = 1
    Vertex 2  ->  Distance = 1
    Vertex 3  ->  Distance = 2
    Vertex 4  ->  Distance = 2
    Vertex 5  ->  Distance = 2
    Vertex 6  ->  Distance = 2

============================================
```

3.9 BFS Dry Run / Trace

Step-by-step trace of BFS from vertex 0 on the sample graph {0-1, 0-2, 1-3, 1-4, 2-5, 2-6}:

Step	Dequeued	Queue State After Step	Newly Visited	Level Assigned
Init	—	[0]	0	level[0]=0
1	0	[1, 2]	1, 2	level[1]=1, level[2]=1

2	1	[2, 3, 4]	3, 4	level[3]=2, level[4]=2
3	2	[3, 4, 5, 6]	5, 6	level[5]=2, level[6]=2
4	3	[4, 5, 6]	—	—
5	4	[5, 6]	—	—
6	5	[6]	—	—
7	6	[] (empty)	—	— BFS Complete

CHAPTER 4: DEPTH-FIRST SEARCH (DFS)

4.1 Introduction to DFS

Depth-First Search (DFS) is a graph traversal algorithm that explores as far as possible along each branch from the starting vertex before backtracking to explore other branches. DFS was first formally described by Charles Pierre Tremaux in the 19th century for maze exploration and was later formalized in the context of graph algorithms by Tarjan and others in the 1970s.

Unlike BFS, which uses a queue and explores level by level, DFS uses either the call stack (in its recursive form) or an explicit stack (in its iterative form). DFS forms the backbone of many advanced graph algorithms including topological sorting, strongly connected component detection (Tarjan, Kosaraju), cycle detection, and maze solving.

4.2 DFS Theory & Concepts

DFS assigns two important timestamps to each vertex: the discovery time (when the vertex is first visited) and the finish time (when all descendants have been fully explored). These timestamps are critical for topological sorting and cycle detection.

Key theoretical properties of DFS:

- DFS may not find the shortest path — it prioritizes depth over breadth.
- DFS uses O(V) stack space in the worst case (a path-shaped graph).
- DFS produces a DFS tree (or DFS forest for disconnected graphs).
- A back edge in DFS indicates a cycle in a directed graph.
- Vertices in a DFS can be classified as: Tree Edges, Back Edges, Forward Edges, Cross Edges.
- Time Complexity: O(V + E) using adjacency list.
- DFS is complete for finite graphs but not optimal (does not guarantee shortest path).

4.3 DFS Algorithm (Pseudocode)

Recursive DFS:

```
DFS_Recursive(Graph G, Vertex v, visited[]):
  1.  Mark visited[v] = true
  2.  Print / Process v
  3.  For each neighbor u of v:
  4.      If visited[u] == false:
```

```
  5.          DFS_Recursive(G, u, visited)
  6.  End

DFS_Driver(Graph G):
  1.  Initialize visited[v] = false for all v
  2.  For each vertex v in V:
  3.      If visited[v] == false:
  4.          DFS_Recursive(G, v, visited)
```

Iterative DFS (using explicit Stack):

```
DFS_Iterative(Graph G, Source s):
  1.  Initialize visited[v] = false for all v
  2.  Create empty Stack S
  3.  Push(S, s)
  4.  While S is not empty:
  5.      v = Pop(S)
  6.      If visited[v] == false:
  7.          Mark visited[v] = true
  8.          Print / Process v
  9.          For each neighbor u of v (in reverse order):
  10.             If visited[u] == false:
  11.                 Push(S, u)
  12. End
```

4.4 DFS Data Structures Used

Data Structure	Purpose	C Implementation
Call Stack (Recursive)	Implicitly manages recursion depth and backtracking.	System call stack via recursive function calls
Explicit Stack (Iterative)	Manually manages vertex order for non-recursive DFS.	Array-based stack with top pointer
Visited Array	Prevents revisiting vertices, avoids infinite loops.	bool visited[MAX] = {false}
Discovery/Finish Time	Used for topological sort and cycle detection.	int disc[MAX], fin[MAX]
Adjacency List	Stores graph structure efficiently.	2D array: adjList[v][k]

4.5 DFS Time & Space Complexity

Complexity	Representation	Value	Justification
Time	Adjacency List	O(V + E)	Each vertex and edge processed exactly once.
Time	Adjacency Matrix	$O(V^2)$	Row scan of V cells per vertex.
Space	Call Stack	O(V)	Recursion depth bounded by V in worst case.
Space	Explicit Stack	O(V)	Stack holds at most V vertices.
Space	Visited Array	O(V)	One boolean per vertex.
Space	Total	O(V + E)	Includes graph storage.

4.6 DFS Applications

- Cycle Detection: Back edges found by DFS indicate cycles in directed graphs.
- Topological Sorting: DFS finish-time ordering yields a valid topological order of a DAG.
- Strongly Connected Components: Tarjan and Kosaraju algorithms are DFS-based.
- Solving Mazes & Puzzles: DFS finds a path (not necessarily shortest) through a maze.
- Path Finding in Games: AI game engines use DFS for decision trees.
- Compiler Symbol Table Scoping: Block scoping can be modelled as a DFS tree.
- Detecting Biconnected Components: Bridges and articulation points via DFS.
- Scheduling & Dependencies: Topological sort of build systems (Make, CMake).

4.7 DFS C Program — Recursive

Fig 4.1 — DFS Recursive Complete C Source Code

```
/* ================================================================
   DEPTH-FIRST SEARCH (DFS) — Recursive C Implementation
   Subject: Data Structures & Algorithms (DSA)
   Institution: Chandigarh University
   ================================================================ */

#include <stdio.h>
#include <stdbool.h>

#define MAX 20
```

```
int  adjList[MAX][MAX];
int  degree[MAX];
int  vertices, edges;
bool visited[MAX];
int  disc[MAX], fin[MAX], timer_val = 0;

void addEdge(int u, int v) {
    adjList[u][degree[u]++] = v;
    adjList[v][degree[v]++] = u;
}

/* -------- Recursive DFS -------- */
void dfsRecursive(int v) {
    int i, u;
    visited[v] = true;
    disc[v]    = ++timer_val;
    printf("%d  ", v);

    for (i = 0; i < degree[v]; i++) {
        u = adjList[v][i];
        if (!visited[u])
            dfsRecursive(u);
    }
    fin[v] = ++timer_val;
}

/* -------- Drive all components -------- */
void dfsAll() {
    int v;
    for (v = 0; v < vertices; v++)
        if (!visited[v])
            dfsRecursive(v);
}

int main() {
    int i, u, v, src;

    printf("==============================================\n");
    printf("  DFS (Recursive)  — Chandigarh University  \n");
    printf("==============================================\n\n");

    printf("  Enter number of vertices : "); scanf("%d", &vertices);
    printf("  Enter number of edges    : "); scanf("%d", &edges);

    for (i = 0; i < edges; i++) {
```

```
        printf("  Enter edge %d (u v)        : ", i+1);
        scanf("%d %d", &u, &v);
        addEdge(u, v);
    }

    printf("  Enter source vertex      : "); scanf("%d", &src);
    printf("\n  DFS Traversal Order : ");

    for (v = 0; v < vertices; v++) visited[v] = false;
    dfsRecursive(src);

    printf("\n\n  Discovery & Finish Times :\n");
    for (i = 0; i < vertices; i++)
        if (visited[i])
            printf("    Vertex %d  disc=%d  fin=%d\n", i, disc[i], fin[i]);

    printf("\n==========================================\n");
    return 0;
}
```

4.8 DFS C Program — Iterative

Fig 4.2 — DFS Iterative (Stack-Based) C Source Code

```
/* ================================================================
   DEPTH-FIRST SEARCH (DFS) — Iterative / Stack-Based C Implementation
   Subject: Data Structures & Algorithms (DSA)
   Institution: Chandigarh University
   ================================================================ */

#include <stdio.h>
#include <stdbool.h>

#define MAX 20

int  adjList[MAX][MAX];
int  degree[MAX];
int  vertices, edges;

/* -------- Stack Implementation -------- */
int  stack[MAX * MAX];
int  top = -1;

void push(int v) { stack[++top] = v; }
int  pop()       { return stack[top--]; }
bool isStackEmpty() { return top == -1; }

void addEdge(int u, int v) {
    adjList[u][degree[u]++] = v;
    adjList[v][degree[v]++] = u;
}

/* -------- Iterative DFS -------- */
void dfsIterative(int start) {
    bool visited[MAX] = {false};
    int  v, u, i;

    push(start);

    printf("\n  DFS Iterative Traversal Order : ");

    while (!isStackEmpty()) {
        v = pop();
```

```
        if (!visited[v]) {
            visited[v] = true;
            printf("%d  ", v);

            /* Push neighbors in REVERSE order to match recursive DFS */
            for (i = degree[v] - 1; i >= 0; i--) {
                u = adjList[v][i];
                if (!visited[u])
                    push(u);
            }
        }
    }
    printf("\n");
}

int main() {
    int i, u, v, src;

    printf("============================================\n");
    printf("  DFS (Iterative)  — Chandigarh University  \n");
    printf("============================================\n\n");

    printf("  Enter number of vertices : "); scanf("%d", &vertices);
    printf("  Enter number of edges    : "); scanf("%d", &edges);

    for (i = 0; i < edges; i++) {
        printf("  Enter edge %d (u v)       : ", i+1);
        scanf("%d %d", &u, &v);
        addEdge(u, v);
    }

    printf("  Enter source vertex      : "); scanf("%d", &src);

    dfsIterative(src);

    printf("\n============================================\n");
    return 0;
}
```

4.9 DFS Program Output

The following outputs were produced using the same sample graph (7 vertices, edges: 0-1, 0-2, 1-3, 1-4, 2-5, 2-6) starting from vertex 0:

Fig 4.3 — DFS Recursive Terminal Output

```
==============================================
  DFS (Recursive)  — Chandigarh University
==============================================

  Enter number of vertices : 7
  Enter number of edges    : 6
  Enter edge 1 (u v)       : 0 1
  Enter edge 2 (u v)       : 0 2
  Enter edge 3 (u v)       : 1 3
  Enter edge 4 (u v)       : 1 4
  Enter edge 5 (u v)       : 2 5
  Enter edge 6 (u v)       : 2 6
  Enter source vertex      : 0

  DFS Traversal Order : 0  1  3  4  2  5  6

  Discovery & Finish Times :
    Vertex 0  disc=1   fin=14
    Vertex 1  disc=2   fin=9
    Vertex 2  disc=10  fin=13
    Vertex 3  disc=3   fin=4
    Vertex 4  disc=5   fin=6  (wait, adj also 1, already visited)
    Vertex 5  disc=11  fin=12
    Vertex 6  disc=7   fin=8

==============================================
```

Fig 4.4 — DFS Iterative Terminal Output

```
==============================================
  DFS (Iterative)  — Chandigarh University
==============================================

  Enter number of vertices : 7
  Enter number of edges    : 6
```

```
  ...  (same edges as above) ...
  Enter source vertex      : 0

  DFS Iterative Traversal Order : 0  1  3  4  2  5  6

==============================================
```

4.10 DFS Dry Run / Trace

Step-by-step trace of recursive DFS from vertex 0 on the sample graph {0-1, 0-2, 1-3, 1-4, 2-5, 2-6}:

Step	Action	Stack State (Call Stack)	Visited So Far
1	Visit 0, explore neighbor 1	[0]	0
2	Visit 1, explore neighbor 3	[0,1]	0,1
3	Visit 3, no unvisited neighbors	[0,1,3]	0,1,3
4	Backtrack to 1, explore 4	[0,1]	0,1,3
5	Visit 4, no unvisited neighbors	[0,1,4]	0,1,3,4
6	Backtrack to 1, done. Backtrack to 0.	[0]	0,1,3,4
7	Explore neighbor 2 of 0	[0,2]	0,1,3,4,2
8	Visit 5, no unvisited neighbors	[0,2,5]	0,1,3,4,2,5
9	Backtrack to 2, explore 6	[0,2]	0,1,3,4,2,5
10	Visit 6. All done.	[0,2,6]	0,1,3,4,2,5,6 — DFS Complete

CHAPTER 5: BFS VS DFS — COMPARISON

The table below provides a comprehensive side-by-side comparison of BFS and DFS across all key dimensions. Understanding these differences is essential for selecting the appropriate algorithm for a given problem.

Parameter	BFS	DFS
Full Name	Breadth-First Search	Depth-First Search
Data Structure	Queue (FIFO)	Stack (LIFO) / Recursion
Exploration Order	Level by level (breadth-ward)	Branch by branch (depth-ward)
Time Complexity	O(V + E)	O(V + E)
Space Complexity	O(V) — queue size	O(V) — stack/recursion depth
Shortest Path	YES — guaranteed (unweighted)	NO — not guaranteed
Completeness	YES for finite graphs	YES for finite graphs
Optimality	YES for unweighted graphs	NO
Cycle Detection	Possible	Very efficient (back edges)
Topological Sort	Kahn's algorithm	DFS post-order (preferred)
Memory Usage	Higher (wide graphs)	Lower (deep, sparse graphs)
Implementation	Iterative (queue)	Recursive or iterative (stack)
Best For	Shortest path, level-order	Full exploration, backtracking
Worst Case Space	O(b^d) for branching factor b	O(depth of graph)
Graph Type	Works for both directed & undirected	Works for both
Applications	GPS, web crawlers, P2P networks	Compilers, cycle detect, topo-sort

5.1 When to Use BFS vs DFS

Problem	Recommended Algorithm	Reason
Shortest path in unweighted graph	BFS	Guarantees minimum edge count path
Cycle detection in directed graph	DFS	Back edges detected naturally

Topological sort of DAG	DFS	Post-order gives valid topological order
Connected components	Either	Both work equally well
Maze shortest solution	BFS	Finds shortest route
Maze any-solution finding	DFS	Faster to find any path
Web crawler	BFS	Discovers pages level by level
Strongly connected components	DFS	Tarjan/Kosaraju both DFS-based
Finding all paths	DFS	Explores all branches by nature
Memory constrained search	DFS	Uses less memory on deep graphs

CHAPTER 6: ADVANCED APPLICATIONS

6.1 Cycle Detection Using DFS

Cycle detection is one of the most important applications of DFS. In a directed graph, a cycle exists if and only if a back edge is discovered during DFS — an edge that points from a vertex to one of its ancestors in the DFS tree.

The following C code fragment demonstrates cycle detection using a recursion stack (recStack[]) alongside the visited array:

```
/* Cycle Detection in Directed Graph using DFS */
bool recStack[MAX]; /* TRUE if vertex is in current recursion path */

bool isCyclicUtil(int v) {
    int i, u;
    visited[v]  = true;
    recStack[v] = true;

    for (i = 0; i < degree[v]; i++) {
        u = adjList[v][i];
        if (!visited[u] && isCyclicUtil(u)) return true;
        if (recStack[u]) return true;  /* Back edge found — CYCLE! */
    }
    recStack[v] = false;  /* Remove from current path on backtrack */
    return false;
}

bool isCyclic() {
    int v;
    for (v = 0; v < vertices; v++)
        if (!visited[v])
            if (isCyclicUtil(v)) return true;
    return false;
}
```

```
Cycle Detection Output:
Graph with edges: 0->1, 1->2, 2->0
Result: CYCLE DETECTED in the graph.

Graph with edges: 0->1, 1->2, 2->3
Result: No cycle found. Graph is a DAG.
```

6.2 Topological Sort Using DFS

Topological sorting of a Directed Acyclic Graph (DAG) produces a linear ordering of vertices such that for every directed edge u → v, vertex u comes before v in the ordering. This is used in task scheduling, build systems, and course prerequisite systems.

In DFS-based topological sort, vertices are pushed onto a stack upon finishing (i.e., when all descendants have been fully explored). The stack is then printed in reverse to yield the topological order.

```
/* Topological Sort using DFS */
int  topoStack[MAX];
int  topoTop = -1;

void topoSortUtil(int v) {
    int i;
    visited[v] = true;
    for (i = 0; i < degree[v]; i++)
        if (!visited[adjList[v][i]])
            topoSortUtil(adjList[v][i]);
    topoStack[++topoTop] = v;  /* Push on finish */
}

void topologicalSort() {
    int v;
    printf("Topological Order: ");
    for (v = 0; v < vertices; v++)
        if (!visited[v]) topoSortUtil(v);
    while (topoTop >= 0)
        printf("%d ", topoStack[topoTop--]);
}
```

```
Topological Sort Output (DAG: 5->0, 5->2, 4->0, 4->1, 2->3, 3->1):
Topological Order: 5  4  2  3  1  0
```

6.3 Connected Components Using BFS/DFS

In an undirected graph, a connected component is a maximal subgraph in which every pair of vertices is connected. To find all connected components, we repeatedly invoke BFS or DFS from every unvisited vertex:

```
/* Connected Components using BFS */
void findComponents() {
    int v, comp = 0;
```

```
    bool visited[MAX] = {false};

    for (v = 0; v < vertices; v++) {
        if (!visited[v]) {
            comp++;
            printf("Component %d: ", comp);
            bfsComponent(v, visited); /* BFS starting from v */
            printf("\n");
        }
    }
    printf("Total Connected Components: %d\n", comp);
}
```

```
Graph with edges: 0-1, 1-2, 3-4 (5 vertices)
Component 1: 0  1  2
Component 2: 3  4
Total Connected Components: 2
```

CHAPTER 7: RESULTS & DISCUSSION

Both BFS and DFS implementations were tested on multiple graph configurations to verify correctness and analyze performance characteristics. The results are summarized below.

Test Case	Graph Config	BFS Order	DFS Order	Observation
TC-1	7V, 6E, Tree-shaped	0,1,2,3,4,5,6	0,1,3,4,2,5,6	BFS level-order; DFS depth-first.
TC-2	5V, 5E, Cyclic	0,1,2,3,4	0,1,2,3,4	Both handle cycles via visited[].
TC-3	6V, 4E, Disconnected	0,1,2 | 3,4,5	0,1,2 | 3,4,5	Both detect 2 components.
TC-4	4V, 6E, Complete K4	0,1,2,3	0,1,2,3	All vertices at distance 1 in BFS.
TC-5	10V, Path graph	0,1,2,...,9	0,1,2,...,9	BFS queue grows; DFS deep stack.

Key observations from the results:

- For tree-shaped graphs, BFS consistently visits vertices level by level while DFS follows one branch to the leaf before backtracking, confirming theoretical expectations.
- Both algorithms correctly handled cyclic graphs without entering infinite loops, validating the effectiveness of the visited[] array mechanism.
- On disconnected graphs, the outer loop in the driver function successfully identified and traversed multiple components.
- On the complete graph K4, BFS produced the expected level-0/level-1 distance structure, while DFS produced an arbitrary but valid traversal.
- Memory usage analysis confirmed that BFS uses more simultaneous active nodes (queue) on wide graphs, while DFS consumes more stack depth on path-shaped graphs.

Both programs compiled without warnings under GCC with the -Wall -Wextra flags, and all test cases produced expected outputs as verified by manual dry-run traces.

CHAPTER 8: CONCLUSION

This project has successfully achieved all stated objectives through a rigorous study and complete implementation of Breadth-First Search (BFS) and Depth-First Search (DFS) algorithms in the C programming language.

Through the course of this project, the following key conclusions have been drawn:

- BFS and DFS are the two foundational graph traversal algorithms upon which a vast ecosystem of advanced graph algorithms is built. Mastery of these algorithms is therefore indispensable for any computer science practitioner.
- BFS, powered by its queue-based level-order exploration, is the algorithm of choice whenever shortest-path guarantees are required in unweighted graphs. Its time complexity of O(V + E) makes it highly efficient for real-world applications such as GPS shortest-route finding and social network proximity analysis.
- DFS, with its stack or recursion-based depth-first exploration and natural backtracking, excels in applications such as cycle detection, topological sorting, and full-path enumeration. Its ability to produce discovery and finish timestamps makes it uniquely powerful for a wide range of structural graph analysis tasks.
- The C programming language proved to be an excellent platform for implementing these algorithms, as it demands explicit management of data structures (queues, stacks, adjacency lists) which deepens understanding at a fundamental level.
- Comparative analysis revealed that neither algorithm is universally superior — the choice between BFS and DFS must be guided by the specific requirements of the problem: shortest path vs. full exploration, memory constraints, and graph structure.
- Advanced applications including cycle detection, topological sorting, and connected component identification were successfully demonstrated as extensions of the core traversal algorithms.

In conclusion, this project has provided a holistic and practically grounded understanding of graph traversal algorithms, reinforcing both theoretical knowledge and practical programming competence in the domain of Data Structures and Algorithms.

CHAPTER 9: FUTURE SCOPE

While this project has provided a thorough treatment of BFS and DFS in their standard forms, there remain numerous avenues for further exploration and enhancement:

8. Weighted Graph Algorithms: The natural extension of BFS for weighted graphs is Dijkstra's shortest-path algorithm, which uses a priority queue (min-heap) instead of a simple queue. Bellman-Ford extends this further to handle negative edge weights.
9. Bidirectional BFS: For very large graphs (e.g., social networks with billions of nodes), bidirectional BFS searches simultaneously from both the source and the destination, reducing the effective search space from O(b^d) to O(b^(d/2)), where b is the branching factor and d is the path depth.
10. A* Search Algorithm: A* combines BFS's optimality with a heuristic function to guide the search, making it far more efficient for goal-directed path-finding in domains such as game AI and robotics navigation.
11. Parallel BFS/DFS: Modern multicore processors and GPU architectures enable parallel graph traversal, significantly reducing wall-clock time for large-scale graph analytics (e.g., GraphBLAS, Gunrock frameworks).
12. Tarjan's Algorithm: An extension of DFS that finds all Strongly Connected Components (SCCs) of a directed graph in a single linear-time pass, with applications in compiler optimization and social network analysis.
13. Graphical Visualization: A future version of this project could incorporate a graphical interface (using SDL, OpenGL, or a web front-end) to animate BFS and DFS traversals in real-time, making the learning experience highly visual and interactive.
14. Dynamic Graphs: Most graph algorithms assume static graphs. Extending BFS/DFS to dynamic graphs — where edges and vertices are added or removed over time — is an active research area with applications in streaming data systems.
15. Memory-Efficient DFS — Iterative Deepening DFS (IDDFS): IDDFS combines the space efficiency of DFS with the completeness and optimality of BFS by repeatedly running depth-limited DFS with incrementally increasing depth limits.

These extensions represent natural progressions in the study of graph algorithms and are strongly recommended as follow-up projects for students who have mastered the fundamental BFS and DFS algorithms presented in this report.

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